

INFORM

A blurred background image of a factory interior. In the foreground, a worker wearing glasses and a grey shirt is looking intently at a piece of machinery. The background shows various industrial components, including pipes and structural elements, in a slightly out-of-focus state.

EXCITING TIMES.

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"SEIZE OPPORTUNITIES."



Dear Reader

The first half of 2021 will soon be over. The Covid-19 pandemic and its associated restrictions continue to affect our day-to-day business, and it remains unclear when we will be able to fully return to normal. But my balance sheet is not all negative. The limitations we faced motivated us to find other ways to offer our customers our usual high level of support. Based on the feedback we've received, it appears that we have mastered this challenge well.

Over these past few months, we have been examining our organization and our product range, and we have made some significant moves. We are convinced the results give us more stability in our day-to-day business and exciting new business prospects. Our new competence area focuses on serving customers in the pharmaceutical and medtech sectors with even greater focus. And, in addition, we can now offer our moulds to PET preform producers who use side-entry platforms. Both these strategic steps were made possible by three key factors in our favour: our almost 70 years of experience in toolmaking, our continuous investment and development, and our loyal, highly capable and highly motivated employees. I consider it a great privilege to have taken these steps with Otto Hofstetter AG. In this issue of INFORM, you can read more about our new MED unit and our side-entry moulds, and you will also meet a valued member of our team.

The two customer portraits we present in this issue show how closely we cooperate with our customers around the world, even when travel is impossible. Sunpet Ambalaj produces high-quality PET preforms in Gaziantep, Turkey, near the Syrian border, and has another plant in Jordan. AYVA Packaging is at home in Sydney and Melbourne, Australia, and supplies the whole country with innovative high-quality cartridges that are produced with our moulds.

Switzerland is always good for a surprising technical innovation. This is borne out by our report on Yvan Bourgnon and his innovative oceangoing catamaran, Manta, which collects waste plastic floating in the seas, recycles it and converts some of it into fuel to power the vessel. All in all, we think this issue of INFORM offers a lot of interesting reading and we hope you will agree. I wish you an enjoyable read and I hope to be able to meet with you in person again soon.

Yours, Otto Hofstetter

Imprint.

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Sunpet Ambalaj started producing PET preforms in Gaziantep, Turkey, in 1998 and established a plant in Jordan in 2010. **Vice President Cevdet Tataroğlu** explains in an interview how the company became **one of the leading PET preform producers** in Turkey and what the future holds for the company.

Under Cevdet Tataroğlu's leadership, Sunpet Ambalaj became one of the top plastic packaging manufacturers in Turkey and Jordan.

You founded your business in 1998. What did your company look like back then and how has it grown since?

In 1998, we started with just one production line in an 8000-square-metre factory. We had three sets of moulds with a total of 48 cavities. Today, we have six injection mould production lines with 10 sets of moulds (with 96 and 128 cavities) operating in a 35,000-square-metre facility, and we also have an 18,000-square-metre factory in Jordan with six production lines and 12 sets of moulds (with 96 and 128 cavities).

We began investing to increase our production capacity in 2005. That's when we started working with a mould manufacturer from Switzerland called OHAG, Otto Hofstetter AG. We also made a big investment in our flexible packaging production in 2015, about 30 million euros. All of our extrusion and printing equipment is from a well-known German machine supplier. Our laminating and production machines come from Italy and our slitting machines are German. Today, we produce packaging for food products like crisps, pasta,

yeast, sauces, coffee and tea, and for non-food products like nappies, wet wipes, detergents and household paper products. And, of course, we make PET preform bottles for water, carbonated drinks and edible oil. Today, after all our growth and investments, we have around 250 employees in Turkey and another 50 in Jordan.

What were your biggest challenges in achieving this significant growth?

We always targeted increasing our production capacity and selling our products to different markets and regions. Of course, our markets have been disrupted by various crises from time to time, but we have come out of them stronger and more confident. Whatever we produce and wherever we sell it, there will be always competition. We choose to compete by maintaining our high levels of reliability, product quality and after-sales service for our customers, and this approach has worked well for us. It's our experience that when our clients win, we win too.



**"MARKETS HAVE BEEN
DISRUPTED BY VARIOUS CRISES,
BUT WE CAME OUT STRONGER
AND MORE CONFIDENT EACH TIME."**

Cevdet Tataroğlu, Vice President of Sunpet Ambalaj

Sunpet makes packaging for food and non-food products, and you make performs. What's the breakdown of production volume between the areas?

Here in Turkey, 50 % of our OHAG preforms are for carbonated soft drink (CSD) products, 45 % are for water and 5 % for edible oils. In Jordan, our OHAG preforms are 60 % CSD and 40 % water. In both locations, there are strong seasonal fluctuations in demand, so production flexibility is an important factor for us. Our moulds from Otto Hofstetter help us efficiently manage these shifts in the production mix throughout the year.

You have said your goal is to become one of the leading companies in the industry. Where do you stand today?

I never targeted having the biggest production capacity. My aim has always been to sell high quality products. We believe, and our experience confirms this, that if we produce good products, our customers will pay a fair price for them. But to answer your question, we are currently the third- or fourth-biggest preform producer in Turkey and the biggest in Jordan.

Continuity is one of your company's core values, one that is highly valued in today's business world. How do you ensure your continuity?

All of our mould system operators, our managers and our quality team have been with us since we started 23 years ago in Turkey and 10 years ago in Jordan. All of our operators are highly trained and very experienced, so we can ensure steady production even when challenges arise.

You have many well-known brands on your client list. Why do you think they choose to partner with you?

I think our excellent relations with the big brands is based on our simple approach to business. We always keep our promises to our customers, whether they are big or small. We deliver good quality products on time. The big brands know they can rely on us and the smaller companies also trust us to deliver what we promise. That formula has been working for us for a long time now and we intend to keep it going.

Sunpet strives for continuous improvement in quality, efficiency and profitability. What actions are you taking to achieve these goals?

We follow the changing needs of the market and invest accordingly. We do not wait for our customers to knock on a competitor's door just because they need one product that we don't make. In other words, we always strive to satisfy the complete SKU needs of our customers. Another big competitive advantage we enjoy comes our financial strength. This



means, for example, that when we think the time is right to buy raw materials, we purchase enough supplies for five or six months of production. In this way, we keep our costs down and we are able to supply a wide range of PET preforms at very competitive prices. This helps us and our customers to be more profitable. A classic win-win outcome.

Your plant is in Gaziantep, in south-central Turkey, near the Syrian border. How has your location influenced your expansion into Jordan?

We have been exporting to countries in the Middle East since 2005. In 2009, some countries in the region increased their customs duties on Turkish-origin preforms. In response, we decided not to exit this market and abandon our customers. Instead, we invested in our second preform factory, in Jordan, in operation since 2010.

"WE'VE TRAINED ALL OUR OPERATORS OURSELVES. THIS WAY THEY ARE USED TO OUR STANDARDS AND CULTURE."

Cevdet Tataroğlu, Vice President of Sunpet Ambalaj

You have set high standards for quality and technology. How do you find the necessary skilled workers for your factories?

We hired two very experienced technicians in 1998 and, since then, we have trained all our operators ourselves. This way, they are all trained to meet our standards and follow our established practices. And, more than that, they are introduced to our culture as a company. We like this approach to bringing in new staff. It's difficult to change someone's habits and routines that may not match our own. That's why we prefer to train our technicians ourselves. >

Sustainability is a priority for your company. You have stated that you aim to have 50% of your packaging recyclable by 2023. What is the motivation to achieve this ambitious target?

Sustainability is no longer just a trend or a fashion. I think we all see now that we have to make a big effort to protect the natural environment. And as a plastics manufacturer, we recognise that, if we don't come up with solutions, then governments and regulators will step in and that will probably be unhelpful for our business. So, for us, sustainability is a challenge that we accept. In this effort, we will, for example, use recycled PET resin for some of our production. We are also working on projects to use scrap from preforms and bottles to be converted into different products. It's the future of our industry and we want to lead here, too.

Sunpet regularly plants trees. Where do they do this?

We own property in various locations. We plant trees on our land every year. We have also been generating some of our energy ourselves. We have invested in a solar energy system, with an array of panels on the roof of our factory in Turkey that has the capacity of 4500 kW/h.

Your many certificates confirm Sunpet's high quality standards. What concrete actions do you take to ensure quality on a day-to-day basis?

We have BRGS and ISO quality management certificates in Turkey and in Jordan. To be able to increase the quality of our products and services, qualifying for these certificates is a very important goal for us.

To what extent do your high quality standards play a role in your choice of moulds?

First of all, it's a basic fact that the moulds we use must be high quality. They should be manufactured to the proper tolerances and they must be delivered and put in service on time. With these expectations in mind, we have repeatedly chosen OHAG as our partner for PET preform injection moulds. Of course, I have to say that OHAG's sales and after-sales teams and Mr Hofstetter himself have always been extremely helpful and friendly. With OHAG, we feel like we are at home.

What other arguments are there in favour of the Swiss manufacturer's products?

Markets are always changing, and meeting the demand for new products is how we secure the future of our business. With Otto Hofstetter, we have a valued partner in developing and enlarging our product portfolio. It's very helpful for us to have a good relationship with our injection mould supplier.

And Otto Hofstetter's after-sales support in areas like process optimisation and on-site training is a real plus for us. Finally, the easy, uncomplicated contact we have with Otto Hofstetter's sales team, technicians and the management makes doing business with them a pleasure. If we need something, we know we can just pick up the phone and a helpful answer will soon follow.

What do you see as the biggest challenges for your company over the next 5 to 10 years?

As I said, markets change. Ours too, and quickly. Consumers' behaviour changes and so do their activities. We have to foresee these changes and be ready to offer our clients the products they need. Today, we're sure about one thing: in the future, flexible production capabilities will be important. And we need to plan our investments accordingly, taking care to give ourselves the capacity and the flexibility that we will need as we grow.

What kind of support do you expect from Otto Hofstetter on this journey?

Otto Hofstetter's deep technical knowledge is a valuable resource for us as we look to develop and expand our current production capacities. We can benefit from their know-how, but we can also benefit from their ingenuity. They seem to like challenges, and that's an attitude we appreciate as we look for ways to meet changing market demands.

**"OTTO HOFSTETTER'S DEEP
TECHNICAL KNOWLEDGE IS A
VALUABLE RESOURCE TO DEVELOP
AND EXPAND OUR CAPACITIES."**

Cevdet Tataroğlu, Vice President of Sunpet Ambalaj

In order to secure its long-term future, what are the main challenges the plastics packaging industry needs to solve?

I know this is a word that we're all hearing every day, more and more, but sustainability, and how we in the plastics packaging industry achieve it, are the twin challenges we face. I see two aspects to this challenge. One, we will surely be seeing more alternative packaging products trying to enter the market in the coming years. Whether plant-based or something else, there will be strong, serious efforts to displace plastic packaging. Some of these alternatives may prove viable, others not. But we have to be ready. We have to be able to point out the many advantages of plastic packaging in terms of product safety, stability and durability, for example. And this leads to the

second part of the sustainability challenge. We have to improve our recycling efforts, from manufacturers like Sunpet all the way to the end consumers. And I'm convinced we can do this. Indeed, we must do this. It's an exciting time to be in this busi-

ness and to be part of the solution to this important challenge!

Mr Tataroğlu, thank you very much for your time and for your thoughtful answers.



Sunpet Ambalaj
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High-quality food and non-food packaging, PET preforms for oil, water and CSD bottles

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State-of-the-art technology and in-house trained operators ensure the highest quality of products for Sunpet's customers.

ROTATED 90 DEGREES.

For over 40 years, **Otto Hofstetter** and **Netstal** have been setting benchmarks in the top-entry segment of PET preform production. With the development of a **side-entry mould** for the new PET-Line series, which features **lateral robot removal**, manufacturers now have the choice.

In 1977, Otto Hofstetter AG built the first injection mould for the production of PET preforms. The client was the Continental Can Company, which had the mandate to produce new drinks packaging for Coca-Cola and Pepsi. The order consisted of 12 injection moulds with 16 cavities, many of which were designed to produce preforms for 2-litre bottles.

THE SIDE-ENTRY WORLD CAN NOW
ALSO RELY ON THE SWISS SUCCESS
TEAM IN THE PRODUCTION OF PET
PREFORMS.

Quality in all dimensions.

In the meantime, well-known PET producers in Asia, North and South America, Australia, Africa and Europe count on the performance of the Swiss-made tools. To achieve

mould design has a much longer service life than conventional feed-in designs, and revisions have been needed after 10 million cycles at the earliest. According to the latest reports from producers, this target can even be exceeded. Other feedback praises the compatibility of the moulds, which significantly increases flexibility when producing lower volumes. For example, in Austria, a 48-cavity preform mould for 1.5-litre bottles has been running without a rebuild for 22 years, and, in Indonesia, a 96-cavity mould for 0.6-litre bottles has been in continuous service since 2007. The hot runners are also marathon performers. In Belgium, one of our hot runners has been in daily use since 2014, clocking almost 16 million cycles and counting.

A new chapter.

Recently, Netstal, our long-standing Swiss machine-building partner, has developed the new PET-Line series with integrated lateral robotic removal and post-cooling. In order to continue this Swiss success story, the manage-

ment of Otto Hofstetter AG decided to build the appropriate side-entry moulds for PET-Line injection units. Thus, our customers can now choose between top- and side-entry layouts, both from Otto Hofstetter AG.

Small twist, same performance.

The move from our proven top-entry to side-entry moulds basically just required a rotation of 90 degrees. For this purpose, some small technical adjustments had to be made in the design. This means that the side-entry moulds include all the quality and features of our top-entry injection moulds. The stack structure is also executed with Cavity-Lock, and the new performance-enhancing options such as OH XCool® and OH Xblow® can also be integrated into the mould. In addition, all end-of-arm-tools (EOATs) are available to match the selected preform. The hot runners are developed and manufactured at Otto Hofstetter AG in Switzerland and comply with all relevant standards.

Impressive test results.

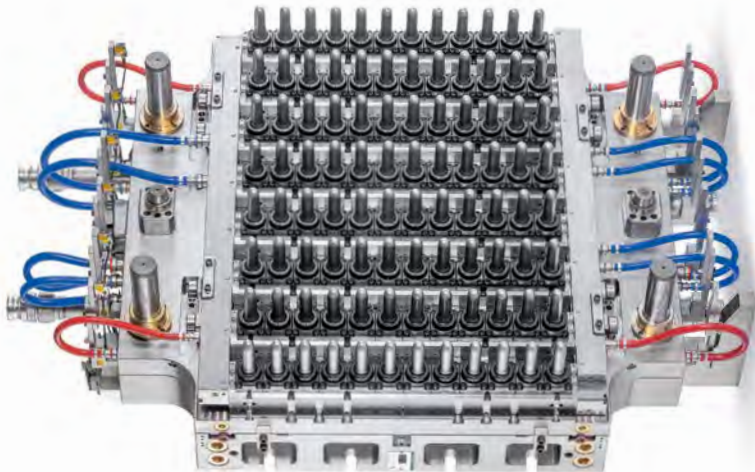
At the beginning of March, a side-entry tool from the Uznach toolmakers was installed in a new PET-Line 4000 for testing purposes at Netstal. The 96-cavity mould was designed for a preform with a 38-millimetre 2-start thread, a weight of 50 grams and a wall thickness of four millimetres. The EOAT cooling ran over four positions, the air cooling over three. After a short break-in run for the mould and some small optimisations, peak values were quickly achieved. With a cycle time of 13.9 seconds and an injection speed of 22.5 grams per second and cavity, Otto Hofstetter AG's new side-entry mould already exceeded expectations in its first tests.

mould availability of more than 95 percent starts with only using high-quality components and having them assembled and checked with great care by highly experienced toolmakers. In addition to uncompromising craftsmanship, to achieve the much-praised performance of the moulds from Uznach requires more. For example, innovative refinements like the OH Xtreme® coating, which enabled a 72-cavity mould to make more than 15 million cycles before its first maintenance. Or OH Xtra® surface treatment, which has reduced average roughness (RA) from 0.2 to 0.3 microns conventionally to only 0.05 to 0.1 microns.

Convincing performance for customers.

As their testimonials confirm, many well-known PET preform manufacturers that Otto Hofstetter AG's Cavity-Lock

MOULD	POST-MOULD COOLING	
	CAVITIES: 96	EOAT-COOL: 4 POSITIONS
	PITCH: 60 X 140 MM	AIR COOLING: 3 POSITIONS
PREFORM	PERFORMANCE	
	NECK: 38 MM 2-START THREAD	CYCLE TIME: 13,9 S
	WEIGHT: 50 G	INJECTION:
	WALL THICKNESS: 4,0 MM	22,5 G/S/CAVITY
MATERIAL		
PET-IV 0,80 DL/G		



Scan the code and see the mould in operation

PUSHING BOUNDARIES.

AYVA Packaging is a privately owned and operated **plastic packaging company in Australia**. It specialises in the production and supply of high-quality plastic pails, barrels and cartridges. **Owner and CEO Michael Schmaedeke** explains in an interview why AYVA relies on Otto Hofstetter's premium moulds and how he aims to push various boundaries with his company in the future.

Michael Schmaedeke bought the company from his parents in 2001 and made AYVA Packaging a leader in the production and supply of high-quality pails, barrels and cartridges in Australia.

You ordered your first mould from Otto Hofstetter – a 16-cavity cartridge mould – by phone. Usually, the sales process involves several steps over the course of a few weeks. What factors assured you that you were ordering the right product?

In 2001 I saw Otto Hofstetter moulds in operation over an extended period of time. You could see the quality of the moulds just by looking at them, but Hofstetter's worldwide reputation, particularly for cartridge moulds, was also a very powerful factor in our decision. During our research, we looked at local Australian mould suppliers but we found that Otto Hofstetter offered the best solutions for our needs.

When was your company founded, and how does it look today?

My father and mother established the company in 1983. I joined in 2001 when my dad became ill. It was supposed to be a temporary job for me for until we found a buyer for the company, but I soon saw opportunities in the business and I took up the challenge of running it. I bought the firm in 2001 and by 2007 our business had increased eightfold. When I started, it was a small company. We had a staff of seven, and our key technician soon left us because he was worried about the future of the business when my dad stopped coming in. So, I had to learn a lot of things very quickly: how production works, how a mould is built, how to run the machines, how to adjust production, how to change moulds, etc. Plus, I had to get to know our customers and the market as a whole.

We were doing very well by 2007, but then came the financial crisis, which set us back a little bit. But it also gave us the opportunity to move into pail products, and that's how the packaging business started, in 2009. I started it from scratch and today that's basically all we do.

"WE HAVE A LOT OF CAPACITY AND WE ARE LOOKING TO EXPAND EXPORTS TO THE REST OF THE WORLD."

Michael Schmaedeke, Owner and CEO of AYVA Packaging

We are now dedicated to packaging products and anything in the packaging field. Today, we're almost 40 times bigger than when we started in 2001. We have factories in two locations and warehouses in four states. We have some ambitious growth targets for the next 10 years. We will be moving to

new facilities later this year, with additional machines, and very soon we will be talking to Otto Hofstetter AG about more moulds as well.

"WE DON'T ENTER A MARKET SEGMENT UNLESS WE SEE AN OPPORTUNITY TO DO THINGS BETTER THAN THE OTHERS."

Michael Schmaedeke, Owner and CEO of AYVA Packaging

At the moment, you are focusing on the Australian market. Are there plans to export to other markets?

Recently we gained some export experience with cartridges to Asia and New Zealand. We have a lot of capacity and we are looking to expand exports to the rest of the world. We are convinced that export markets will welcome what we have to offer. Like Switzerland, Australia is not necessarily a low-cost country, but we do have strict quality controls, and, with our innovative IML decorated cartridge, the demand is there for our products in other regions like Asia and the US.



The range of plastic packaging solutions is very broad. Why did you choose to focus on pails, barrels and cartridges?

Our growth strategy is to continue to find new markets and new products. Four or five years ago, we were driven to cartridges by customer demand. We don't enter a market segment unless we see an opportunity to do things better than what is already available. Cartridges were ready for improvements, for example, with in-mould labelling decoration, and we felt that we could do that. There was strong demand in the domestic market that supported our focus on that segment.

>

"OTTO HOFSTETTER'S SUPPORT DURING THE DEVELOPMENT OF OUR CARTRIDGE PRODUCT HELPED US SUCCEED."

Michael Schmaedeke, Owner and CEO of AYVA Packaging



Your website claims that Ayva is an industry-leading source for plastic packaging products. How do you fulfil this promise?

We achieve it through innovation and continuous improvement. We introduced decorated cartridges to the market here. That's one example. We also brought a one-piece grease cartridge to the Australian market with in-mould labelling that provides excellent decoration. In addition, our customers get the quality of products made with Otto Hofstetter's moulds, which means we can deliver uniform mould sections. In the pails business, all of our products are unique designs, and with each new product we bring a new feature to the market and improve their performance. We listen to our customers and we take their feedback as an opportunity to improve our products. We also create new products to solve problems that we hear about from our clients.

To what extent do your high quality standards play a role in the choice of Otto Hofstetter moulds?

Manufacturing is difficult enough, so you have to start with good moulds and good machines. Only then can you guarantee the quality of your products.

From a producer's point of view, how would you describe the advantages of Otto Hofstetter moulds?

Let's look at moulds for cartridges, for example. Otto Hofstetter are specialists in that field. They create, develop and continuously improve the design of their moulds to ensure that mould sections are uniform and mouldings are straight. They have years of experience in making cartridges, and that know-how has really helped us to succeed in the market.

How is it to work with Otto Hofstetter, who are based in Switzerland, with you in Australia? That's halfway around the world.

The cooperation has been very good. For us, the quality of the service we get is as important as the quality of the product we buy. Fortunately, we haven't had many problems with Otto Hofstetter equipment, and the support from Otto Hofstetter's technical team has been excellent. And, when we tell them that we want to develop something new or improve an existing product, they come up with good, feasible ideas. They are valued discussion partners. We also work with a local tool-maker here in Australia who has a good relationship with Otto Hofstetter. This is important because we can't afford the time

and expense of sending moulds back to Switzerland every time some work is needed. The support Hofstetter provides our local toolmaker is a big plus for us.

You strive to deliver cost-efficient solutions to your customers. How do you meet this goal when working with tools from high-priced Switzerland?

Otto Hofstetter moulds are substantially more expensive, initially. That's true. But when we look at the overall service life costs, we believe that the speed, efficiency, low generated waste levels and the lightweight products we make more than justify a high initial cost. Otto Hofstetter moulds last a long time, and that certainly has to be part of the cost calculation. To me, Otto Hofstetter's quality make them the right decision commercially over the long term.

Your entire product range is 100% recyclable. How did you manage to do this?

With cartridges, for example, by having in-mould labelling, we use uniform materials in the whole product, for the nozzle, the cylinder and the labelling, which makes it simple to recycle the cartridges.



Sustainability is an important topic for you. Which measures do you already have in place and what is planned for the future?

We reduced the size of some items so that they can be more easily returned to recycling stations, and our production facility uses energy-efficient machines. Also, we invest in good machinery. Efficiency reduces waste. We also make an effort to minimize and separate our waste. We are cooperating with our suppliers as well to ensure that they have the same philosophy.

"FOR US, THE QUALITY OF THE SERVICE WE GET IS AS IMPORTANT AS THE QUALITY OF THE EQUIPMENT WE BUY."

Michael Schmaedeke, Owner and CEO of AYVA Packaging

Can Otto Hofstetter moulds help you reduce the amount of materials you use and make your production more sustainable?

Making products lighter is the best way to reduce the consumption of materials. That's where Otto Hofstetter comes in. For one thing, we have to gain market acceptance for lighter-weight products, and we're working with our customers on this. And OHAG's support in those projects is crucial to their success.

With cartridges, unlike with PET bottles, it's difficult to reduce materials because of the risk of deformation. How are you meeting this challenge?

That's right. If you have cartridges that are too thin, they can fail. The environmental costs of leaked silicon sealer or machine grease would far exceed the savings gained from using less material to manufacture the cartridge. Technology is not an end in itself. Otto Hofstetter moulds help ensure uniform mould sections, which allows us to offer the market a very lightweight cartridge, relatively speaking. But we need to push the boundaries further.

What kind of support are you looking for from Otto Hofstetter in future?

Our relationship is still young. So far, everything has been very positive. When an issue has arisen, response times have been very good. So, I hope they keep doing what they have been doing, with continuous improvement, of course. We are confident that we can partner well with them in future. Their support during the development of our cartridge product was

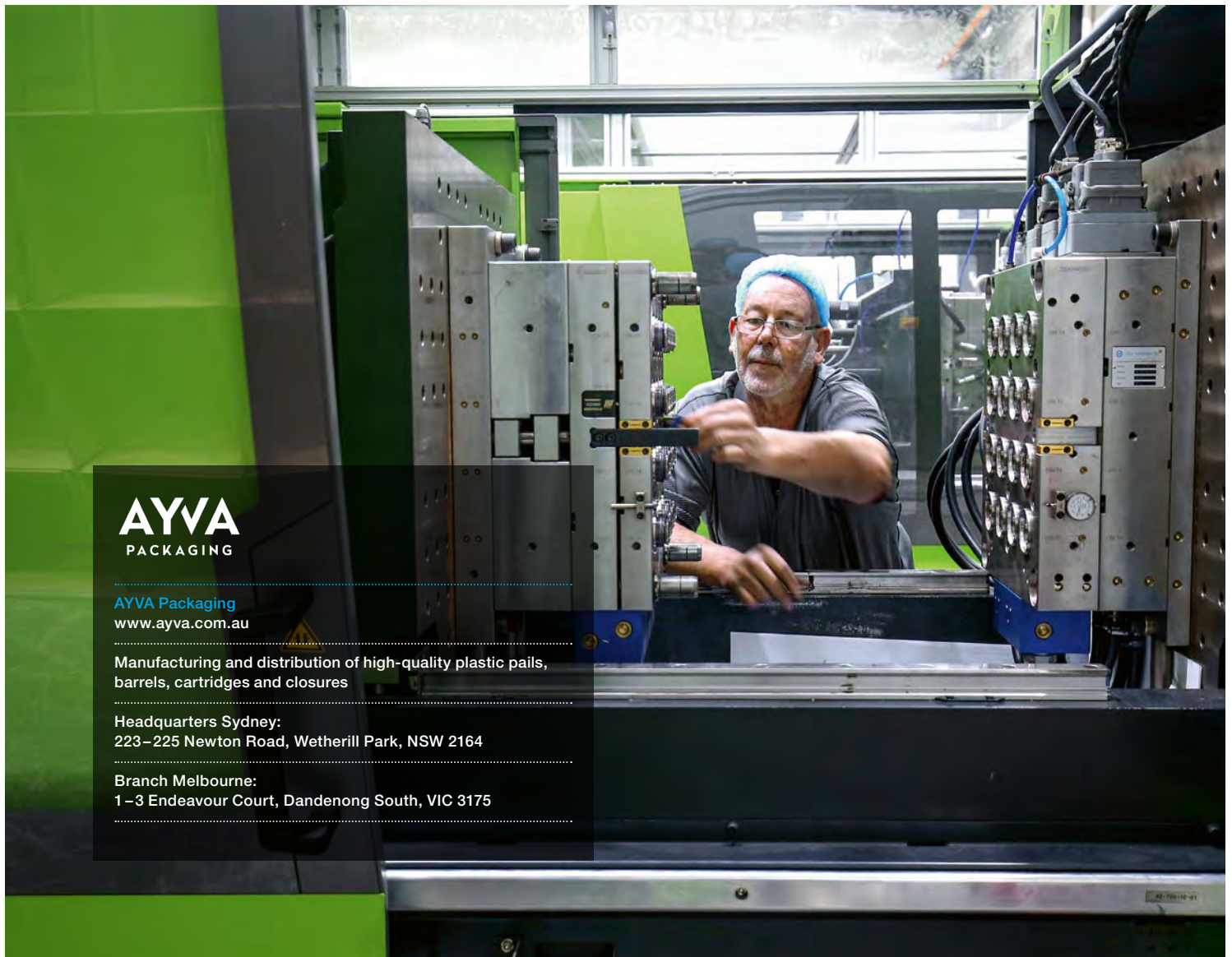
really important and helped us succeed. OHAG put in a lot of extra engineering and development work to make it happen, and that is 100% remembered and appreciated. That's why we can strongly recommend Otto Hofstetter as a supplier.

What do you see as the biggest challenges for your company over the next 5 to 10 years?

I think the biggest challenge that our industry faces is the

negative perception of plastics and the availability of sufficient quality recycled resin. This is a global challenge, but in Australia we really have a lot to do. We believe that if we can improve recycling here in Australia, we will provide a real benefit to our society and to our clients.

Many thanks for the insightful interview and all the best for your future with AYVA.



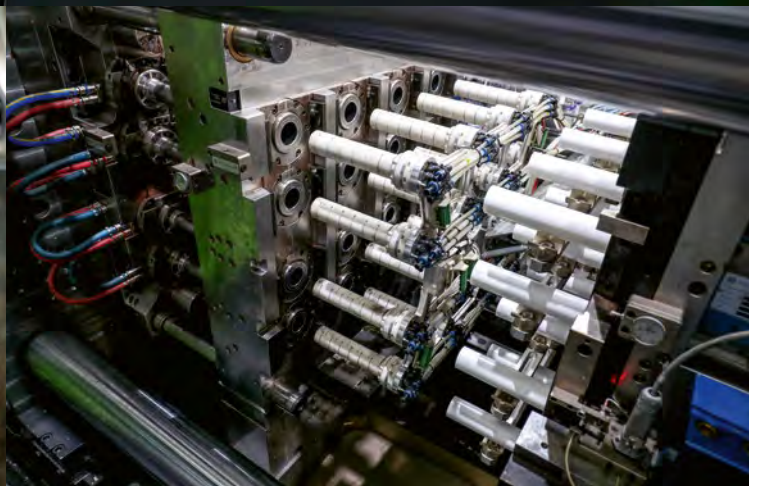
AYVA
PACKAGING

AYVA Packaging
www.ayva.com.au

Manufacturing and distribution of high-quality plastic pails, barrels, cartridges and closures

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Branch Melbourne:
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A LOGICAL STEP.

As of **February 1, 2021**, **Otto Hofstetter AG** expanded its offering. In addition to its traditional competence areas of PAC and PET, the Swiss family business **enters** into **production** of injection moulds for the **pharmaceutical and medtech industries**.

Otto Hofstetter AG first started constructing injection moulds back in 1955. Ever since, thanks to the continuous development of its products and services, this Swiss family business has been a reliable choice for packaging and preform manufacturers around the world.



Continuously developing.

With the introduction of the first injection mould for thin-walled packaging in 1962, Otto Hofstetter AG produced a pioneering innovation for the plastics packaging industry. Numerous additional in-house developments not only created solutions for many new applications, but also increased productivity and reduced total life-cycle costs for injection moulds.

Many world-renowned food and cosmetics brands count on injection moulds made by Otto Hofstetter AG. With its latest development, the S-Line, the Swiss company offers semi-standardised moulds that set a new benchmarks in terms of delivery time, investment volumes and ecological sustainability.

Competencies combined.

Otto Hofstetter AG launched a new MED competence area in February 2021. The MED division focuses on the development and construction of injection moulds for medical and pharmaceutical plastic products. The company has successfully realised various projects for well-known medtech companies over the years. Now, to coordinate the relevant skills, services, processes and quality controls, a new business unit has been launched: OHAG MED.

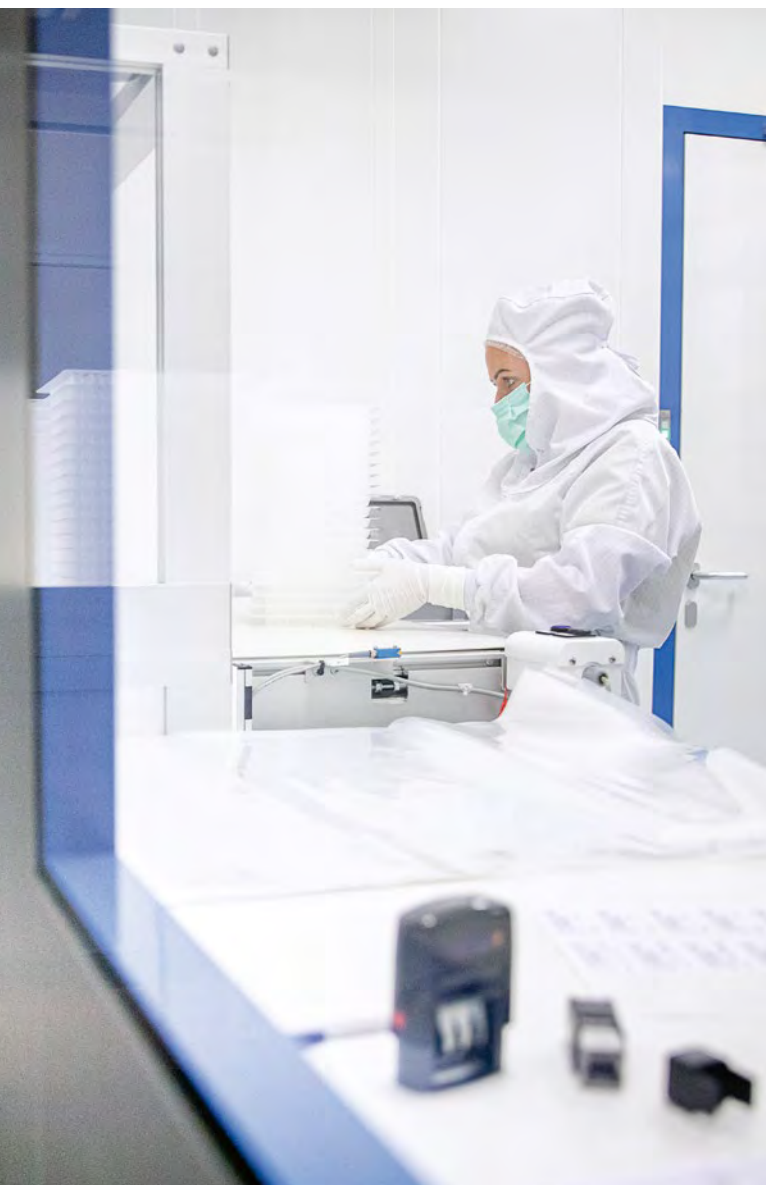


Support from planning to production.

Otto Hofstetter MED designs and manufactures injection moulds for pharmaceutical primary packaging and for medical and diagnostic products, as well as packaging for dietary supplements.

**OHAG MED'S SERVICES, PROCESSES
AND QUALITY ASSURANCE
ARE FINE-TUNED TO MEET TIGHTLY
DEFINED SPECIFICATIONS.**

As with PET and PAC, our services for the MED sector range from consultation to development to product samples under production conditions. We also provide guarantees on moulds for the pharmaceutical and medtech industries. And our after-sales team is ready to support customers anywhere and any time.



SAVIOUR OF THE SEVEN SEAS.



Every minute, 17 tonnes of plastic waste is dumped into the world's rivers, lakes and oceans. Obviously, this dismal state of affairs is a growing and unsustainable burden on the environment. Five years ago, the Swiss adventurer and record-setting sailor Yvan Bourgnon decided he could no longer simply sit back and watch the damage accumulate. In 2016, he founded SeaCleaners, an organisation dedicated to protecting the world's oceans. This spring, SeaCleaners presented its flagship project, Manta, an ocean-going plastics-recycling catamaran.

Yvan Bourgnon has sailed the seven seas for more than 30 years, competing in many regattas as well as sailing just for pleasure. He has celebrated many victories and set numerous records. During his round-the-world navigations in 2013 to 2015, he could not help but notice how quickly the pollution of the seas had increased over the years. This observation motivated him to form SeaCleaners.

A ship with a destination.

With over three years of development time, the club has designed a unique ocean-going catamaran that will “fish” plastic from the seas. The nearly 57-metre-long and 26-metre-wide Manta collects plastic particles a size of one centimetre and larger using three collection nets at the stern of the ship with a combined span of 46 metres. The nets reach to a maximum depth of one metre under the water’s surface and can extract one to three tonnes of plastic waste per hour. That may not sound like a lot, but the planning calls for the ship to be in use 300 days a year. Thus, Manta’s projected annual collection capacity is between 5000 and 10,000 tonnes.

The world’s largest floating recycling facility.

If the catamaran is launched as planned in 2024, it will be the largest floating waste processing plant in the world. The collection nets are emptied on board and the waste enters a sorting system. All plastic waste is fully processed onboard Manta. Metal, glass and various light metals are sorted manu-

ally and stored. Plastic waste is treated with special care. It is sorted and then compressed for storage. Organic materials, such as wood or seaweed, is returned to the sea, as long as the existing marine conditions permit.

**MANTA WILL COLLECT 5000
TO 10,000 TONNES OF
PLASTIC WASTE A YEAR.**

A sophisticated and sustainable propulsion system.

There are already some systems to collect waste from the sea, and even a few ships dedicated to this task. But Manta’s hybrid drive system is unique worldwide, as it generates its energy from renewable sources. It is equipped with sails about 60 metres high, wind turbines and hydropower generators. Much of the deck is covered with photovoltaic solar panels that generate electricity. These are effective, but in themselves not new. What is truly innovative on board Manta is its waste incineration plant. This is the central element in the ship’s energy self-sufficiency. The collected plastic is turned into electricity through a waste-to-electricity conversion unit and the electricity feeds all the equipment onboard. In the conversion, the plastic evaporates and releases a synthetic gas. In turn, this gas drives turbines and generates more

Scan the code and
watch the video.



At 57 metres long and 26 metres wide, Manta will be the largest floating plastic recycling plant in the world.





Switzerland's Yvan Burgnon developed Manta, an innovative ocean-going catamaran, together with his association, SeaCleaners

electricity that is either used immediately or stored in batteries. With this innovative hybrid drive, 75 per cent of Manta's power needs are met autonomously, without fossil fuels.

A laboratory on the high seas.

In addition to removing plastics from the world's oceans, the goal of SeaCleaners is to better understand pollution in order to combat it more efficiently. For this reason, 10 scientists will work on board in dry and wet laboratories. They will concentrate on the geopositioning of waste masses and the quantification and analysis of the plastic pollution. The research data should be available to everyone on an open data platform and, above all, it should be used to raise awareness of the battle against plastic pollution in the oceans.

THE WASTE-TO-ELECTRICITY CONVERSION UNIT SUPPLIES ENERGY FOR MANTA'S EQUIPMENT.

On duty around the world.

The ocean-going recycling catamaran Manta will mainly be used in the waters of Africa and Asia. The 10 rivers with the highest levels of plastics pollution include the Niger, the Yangtze and the Mekong, and, according to recent research, they contribute up to 95 per cent of the plastic pollution in the world's oceans. Over 80 per cent of plastic pollution originates on land and is carried by rivers to the seas and oceans. Manta will patrol the coastal waters and collect the plastic flotsam before it sinks or is rinsed further into the sea and pollutes larger areas. Hurricanes and tsunamis can accelerate the spread of plastic waste. After such events, Manta can be pressed into service to contain the spread of pollution.

Construction can begin.

Manta will only set sail in three years at the earliest. Until then, the various items of equipment and the many components are being extensively tested, and trial expeditions will be made. Manta's definitive equipment list will be compiled on the basis of these trials. In 2023, the construction of the catamaran should begin at an as yet unknown shipyard. After another year, the first systems will be mounted and installed. After launching, the first sea trials will be held. When all systems get the green light, hopefully in 2024, Manta will begin its important fight against plastic pollution in the world's oceans.

TECHNICAL SPECIFICATIONS:

LENGTH: 56.5 METRES, WIDTH: 26 METRES, HEIGHT: 62.5 METRES

COLLECTION CAPACITY PER HOUR: 1 TO 3 TONNES

SIZE OF COLLECTED WASTE: FROM 10 MILLIMETRES

COLLECTION DEPTH: 1 METRE UNDER SURFACE

CREW: 34



MAINTAINING AN OVERVIEW IS EVERYTHING.

With a two-year break, she has been with Otto Hofstetter AG for a total of **19 years**. She's a **design engineer**, and the breadth of her experience makes her the go-to person for **complex tasks and demanding shapes**. By nature, she has many interests, but she likes to maintain a good overview. Perhaps that's why **working from home** suits her so well. Her name: **Manuela Böni**.



As part of our series profiling Otto Hofstetter AG employees, the people who contribute every day to supporting the world-renowned quality of our products and services, this time we offer you a portrait of a member of the engineering department.

After 17 years at Otto Hofstetter AG, Manuela Böni decided she wanted a change. She took a look at other professional options and returned to us two years ago. She works in our traditional field of expertise as a design engineer. At the beginning of her career, she completed a four-year training programme as a mechanical draughtswoman, and she has continued to develop her skills and know-how over the years. Since she lives some distance from Uznach, she has been working from home during the pandemic. She lives in a tranquil region of Switzerland, in the foothills of some impressive mountains, where excellent vineyards can be found. In this showplace of nature, and with yoga as part of her routine, our interviewee provides fresh energy.

Manuela Böni, what exactly is your job at OHAG?

My area of responsibility includes the creation and development of technical drawings for preforms and PET moulds. Thanks to my 19 years of experience, the trickier preform drawings usually end up on my desk. This work is not only technically demanding, it's really about finding the best solutions and working with the technical sales people to meet the customer's specific needs while at the same time ensuring the durability of the equipment and the quality of the end products. This is often a challenge, but as a team we have always found a solution.

**"IT'S ABOUT MEETING THE
CUSTOMER'S NEEDS WITH QUALITY
AND PRACTICALITY."**

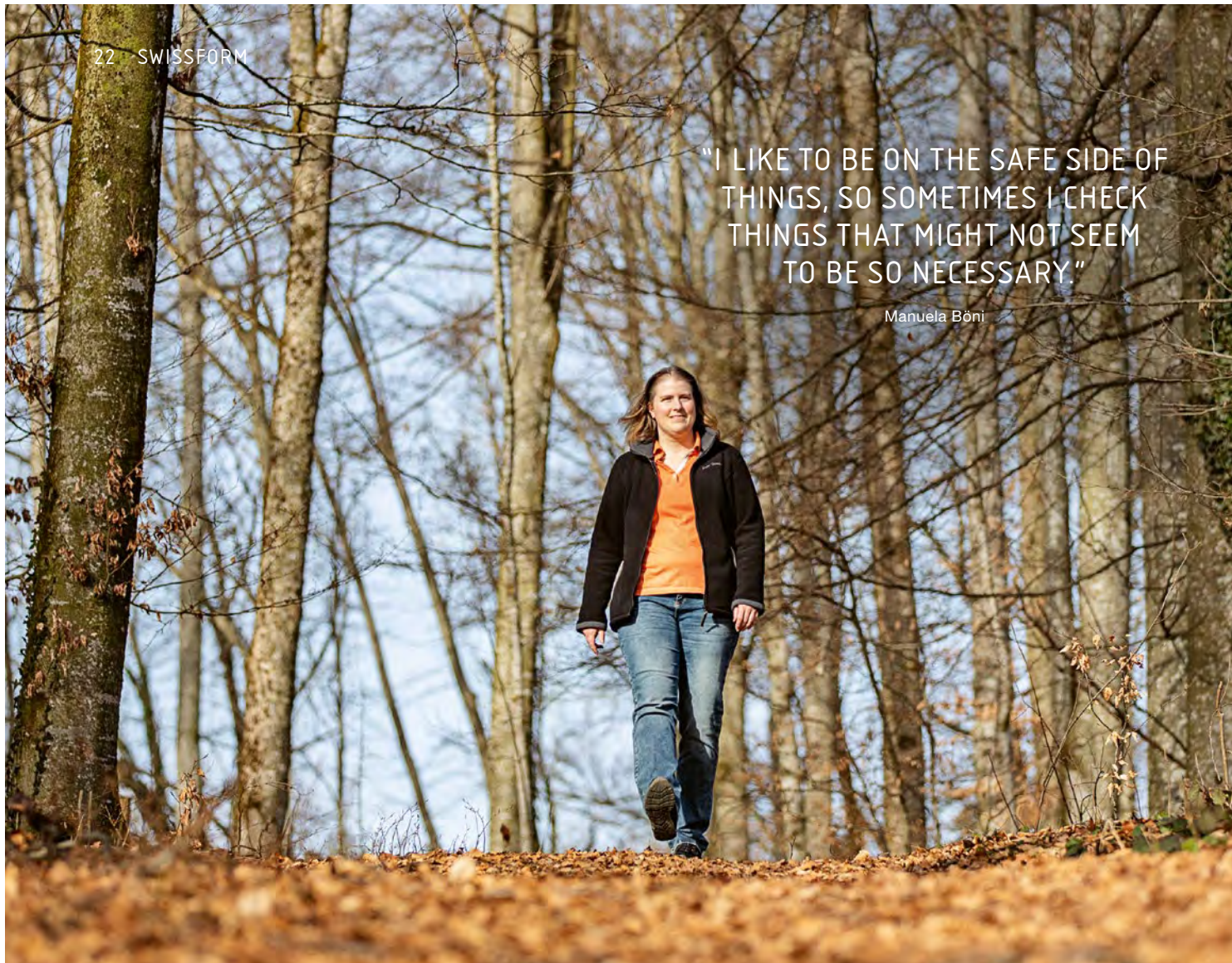
Manuela Böni

**A design engineer working from home is rather unusual.
How does this arrangement work for you?**

For me, this is the perfect arrangement. In Uznach, about 20 people work in the engineering office in one big room. It's natural that, at any given time, something is being discussed somewhere in the room, or a customer calls, or one of us has to vent a bit of emotion if something doesn't go as planned (or indeed if it does!). At home, I am not distracted by anything and I can focus on my tasks 100 percent. If someone has to talk to me, I am reachable via team chat and phone all the

"I LIKE TO BE ON THE SAFE SIDE OF THINGS, SO SOMETIMES I CHECK THINGS THAT MIGHT NOT SEEM TO BE SO NECESSARY"

Manuela Böni



time. So, I'm present but absent, and for me that's ideal. What I sometimes miss is the proximity to the workshop, as well as the casual contact with my colleagues. From time to time, there are technical questions that I'd like to discuss, which we do, of course, online, but it's not quite the same. That's why I work one or more days a week in Uznach.

What is OHAG for you?

After completing my apprenticeship, I began my career at Otto Hofstetter AG. Until the two-year detour to check out other companies, I spent my whole professional life here. I like the work, so I wanted to come back. At Otto Hofstetter AG, it's not all peace, joy and harmony, but it doesn't have to be like that for me, either. I stand by my opinions and openly say what I think. That's possible to do here and it's appreciated by everyone – OK, with maybe a few exceptions. But that also works for me.

What do you look forward to most when you go to work?

Basically, the work itself. Back when I was in school, when we girls were asked to choose between home economics and geometry, the technical side of things interested me more. And, when the time came to choose a career, it was clear for

me from the beginning that I wanted to do something technical. This interest remains, though my technical interests run in many directions, not just preforms or packaging.

What do you do outside of OHAG?

No much technical stuff, except when there is something to repair at home. Otherwise, I like walking in nature. The area where I live is ideal. Mountains, vineyards and a lovely landscape provide plenty of choices for walks. In addition, I like to discuss all kinds of things with people. I'm interested in the opinions of others. I do work intensely, I guess, so to relax, I like to read. Here, too, I have wide interests, except for politics and history. When my 10-year-old godchild recently recommended a book to me from the children's detective series "The Three!!!", I said I'll read it. It's a welcome distraction for me.

You have led us to a small hill at the edge of the forest. Below us is the village of Uznach and in the distance the mountains. Why is this your favourite place?

There are many wonderful places around Uznach. This one is near where I grew up. As children, we spent a lot of time up here. We played in the forest with stones, nuts and berries and

then ate them. We knew exactly where there were berries. In addition, this place was within walking distance of home. So, if someone's mother called out, we could hear it. Later, as a teenager, we girls came here to share our worries and joys.

The view is wonderful. You can see the whole area. To what extent are views and maintaining an overview important to you?

The view from this spot touches me deeply every time I see it. I love to look out over the wide, flat Linth Plain to Lake Zurich. Having an overview is very important to me. On the other hand, I don't like surprises. I like to know what's coming my way and where I stand, whether the situation is positive or negative. That gives me security. In general, I like to be on the safe side, so I check things that might not seem necessary. That's just how I am. Trust is not always easy for me, but I realise you cannot always question everything.

**"TODAY, MY COMPETENCE IS
RECOGNISED BY MEN."**

Manuela Böni

Do you know the names of the mountains we see beyond the Linth Plain?

Should I list them? Seriously, I know them all. And not only that. I've hiked up almost every one of them. With our parents, we often used to drive to the mountains for hikes. There was a phase when I didn't necessarily share my parents' enthusiasm for hiking. But, as I got older, I understood it better. Today I often hike for three or four hours at a time and enjoy every moment.

You grew up in Uznach. What does it mean for you to have roots?

For me, having roots means knowing who you are. It also means having something familiar and without surprises. In terms of Uznach, that's true for me. I was born and raised here. I feel at home here. But that doesn't mean I want to live here. In fact it's the opposite. Uznach has changed a lot and there are other, more pleasant places to live, in my opinion. My parents and my work are here, so I always get my fill of Uznach, and that does me good.

How much are you rooted at OHAG?

When you've been here as long as I have, you know how the

shop runs. You know who you can fully trust and who is best treated with a bit of caution. That was also a major reason why I wanted to return to Otto Hofstetter AG. I knew what to expect and I knew most of my colleagues – people I like and with whom I can work very well together. It was important to me not to just jump into the unknown. Here, I know what's expected of me. I had proven to myself that my skills were also appreciated at other companies. And that strengthened my self-confidence.

What does it take to be comfortable as a woman in a technical profession?

I really got on to the technical track early. In vocational school, I was the only girl. If the boys got into some mischief, at first the teachers hoped I would tell on them. If I had betrayed the troublemakers, they would no longer have accepted me. As an apprentice, a supervisor once gave me a lower grade than the boys for comparable performance. But that's ancient history now. Today my competence is recognised by men. Only very rarely does someone seek a second opinion from a man after I've expressed my views. But I don't get annoyed by that. My energy is too valuable to worry about such things.



How could OHAG improve?

That's a tough question. Sometimes I guess I would like to see somewhat better communication. What's more, I would be grateful to get feedback on performance when it is flawless. It's not about getting a pat on the back after every job. But from time to time you do like to hear: "That was good work." Yes, that would be nice.



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